```
> restart;
  # lagrangian_reconstructed.mpl
  # Reconstructed, runnable Maple source for lagrangian.mw
  # (Michael Vio, Lagrangian QED & Force Between 2 Protons)
  #
  # Regenerated 22 August 2026 by decoding the worksheet's internal
  2-D math
  # format (Typesetting presentation trees, base64/M-format
  encoded) back into
  # 1-D Maple syntax. Verified by re-execution in Maple 2025 (see
  # lagrangian_transcript.md / .docx for the full annotated
  transcript).

  ### Section 1: physical constants ###
  R := 3.567 * 10^(22);
  mp := 1.672621923 * 10^(- 27);
  hbar := (6.62607015)/(2 * Pi) * 10^(- 34);
  r2 := 1.2 * 10^(- 15);
  m := 9.10938356 * 10^(- 31);
  c := 299792458;
  ec := 1.602176 * 10^(- 19);
  h := 6.62607015 * 10^(- 34);
  B2 := 6.596177300 * 10^(12);
  epsilon0 := 8.854187813 * 10^(-12);
  rb := 5.292410932 * 10^(-11);
  re := 2.82 * 10^(- 15);
  rp := 1.2 * 10^(- 15);
  r0 := 4.2 * 10^(- 15);
  hm := h * 4.794614950 * 10^(6);
  hme := 1.983113649 * 10^(-8);
  E0 := 5.747437648 * 10^(-19);
  B1 := 3.910395210;

  ### Section 2: radial gravitational/magnetic model g(r) ###
  2 * 10^(- 64) * R^(2);
  # At radius = 1/R

  g2 := evalf((2 * 10^(- 64))/(r^(2)));
  diff(((2 * 10^(- 64))/(r^(2))), r);
  evalf(- (2 * 2 * 10^(- 64))/((3.566 * 10^(22))^(- 3)));

  g1(r) := - (2 * 2 * 10^(- 64))/(r^(3));
```

```
solve(g1(r) = - R);
r1 := 2.238303432 * 10^(-29);
solve(g1(r) = - R);

# NB: `g` itself is never explicitly (re-)assigned in the source
at this
# point, so the calls below evaluate to the unevaluated symbolic
form g(...)
# -- this matches the original worksheet's own output exactly.
g((25)/(R));
g1((25)/(R));
g(2.238303432 * 10^(-29));
(25)/(R);

f1 := (2 * 10^(- 64))/(( - 7.041546897 * 10^(-22))^(2));
# Value of gravity in Newton at radius of 7*10^(-22) m

r1 := 7 * 10^(- 22);
(2 * 10^(- 64))/(( 7 * 10^(-22))^(2));
evalf((2 * 10^(- 64))/(r1^(2)));
(2 * 10^(- 64))/(r1^(2));
# -(2.015245098 * 10^(-64))/(r^3)

### Section 3: spin angular velocities and tangential speeds ###
omegae := 2 * Pi * ((5 * ec * B2 * hm)/(re^(2) * m))^(0.5);
# angular rotation of electron in radians per second

omegap := 2 * Pi * ((5 * ec * B2 * hm)/(rp^(2) * mp))^(0.5);
# angular rotation of proton in radians per second

omegap * rp;
# tangential velocity at equator of proton

omegae * re;
# tangential velocity at equator of electron

gama := (omegae * re)/(c);

omega2 := ((5 * hm * ec * B2)/(m * re^(2)))^(0.5);
# angular rotation of electron in radians per second due to the
magnetic field of the nucleus
```

```
### Section 4: Coulomb, Lorentz and spin-magnetic force estimates
###
F21 := 2 * Pi * omegap * rp * (2 * 10^(- 64))/(((4.2)/(R))^(3));
2 * Pi * omega2 * re * (2 * 10^(- 64))/(((1)/(R))^(3));
2 * Pi * omegap * rp;
# F21: the magnetic force due to proton spin, ~122 Newton

omega2 * re;
# tangential velocity at electron equator due to the nucleus's
magnetic field

Fes := (ec^(2))/(4 * Pi * epsilon0 * 4 * rp^(2));
# The Coulomb repulsion between the glued protons, ~40 Newton

FLorentz := ec * ((2 * 0.08 * 10^(- 15))/(mp))^(0.5) * B1;
# The Lorentz force generated by the proton moving linearly with
energy
# E1 = 500 eV (0.08e-15 J), speed ~3e6 m/s

Vp := ((2 * 0.08 * 10^(- 15))/(mp))^(0.5);
(0.08 * 10^(- 15))/(1.602 * 10^(- 19));
# The proton velocity Vp at the translation energy of 500 eV

2 * Pi * ((5 * ec * B2 * hm)/(rp^(2) * mp))^(0.5) * (2 * 10^(-
64))/(rp^(2));
(2 * 10^(- 64))/(rp^(3));

### Section 5: time-dependent magnetic-field PDE and closed-form
solution ###
restart;
# Electromag

# NB: the original worksheet writes H(r) as a function of r alone
even
# though `de` also differentiates it in t, so Maple's pdsolve
refuses with
# "the indeterminate function, H(r), depends on only one
variable". H is
# corrected to H(r,t) here so pdsolve can actually solve the PDE.
de := diff(H(r,t) * (1 + exp(1)^(- c * t) * A), t, t)
      - c * diff((H(r,t) * (1 + exp(1)^(- c * t) * A)), r, r) =
0;
```

```
bc := H(2 * R, t) = 0, D[1](H)(1/R, t) = R;

pdsolve(de, H(r,t));
# The general (d'Alembert-type) solution above is what pdsolve
can obtain in
# closed form for arbitrary A, c, R; feeding it the boundary
conditions `bc`
# together with symbolic A, c, R makes pdsolve's internal solver
time out,
# and substituting the worksheet's numeric A=1, c=30, R=35.67
makes it fail
# outright ("unable to isolate the functions ... in the given
boundary
# conditions") because these are r-boundary conditions holding
for all t,
# not the initial/boundary form pdsolve's automatic solver
expects. The
# author evidently applied `bc` by direct ansatz instead --
exactly
# reproducing the general solution's structure -- to reach the
closed-form
# Fmag(r) below.

Fmag := ((sqrt(1 + exp(1)^(- c * t ) * A ) * R * (exp(1)^(- (2 *
(R - (r)/(2)) * sqrt(c) * sqrt(A) * exp(1)^(- (c * t)/(2)))/(sqrt
(1 + exp(1)^(- c * t ) * A ))) - exp(1)^((sqrt(A) * sqrt(c) * exp
(1)^(- (c * t)/(2)) * (2 * R - r))/(sqrt(1 + exp(1)^(- c * t ) *
A )))))/(sqrt(c) * sqrt(A) * (exp(1)^(- (c * t * sqrt(1 + exp(1)^
(- c * t ) * A ) * R - 4 * exp(1)^(- (c * t)/(2)) * (R^(2) - (1)/
(2)) * sqrt(A) * sqrt(c) )/(2 * sqrt(1 + exp(1)^(- c * t ) * A )
* R)) + exp(1)^(- (c * t * sqrt(1 + exp(1)^(- c * t ) * A ) * R +
4 * exp(1)^(- (c * t)/(2)) * (R^(2) - (1)/(2)) * sqrt(A) * sqrt
(c) )/(2 * sqrt(1 + exp(1)^(- c * t ) * A ) * R)))) )/(4 * Pi *
r^(2));

F := subs(A = 1, t = 0, Fmag);

Fmag := simplify((sqrt(2) * R * (exp(1)^(- (2 * (R - (r)/(2)) *
sqrt(c) )/(sqrt(2))) - exp(1)^((sqrt(c) * (2 * R - r))/(sqrt(2)))
))/(4 * sqrt(c) * (exp(1)^((2 * (R^(2) - (1)/(2)) * sqrt(c))/
(sqrt(2) * R)) + exp(1)^(- (2 * (R^(2) - (1)/(2)) * sqrt(c))/
(sqrt(2) * R))) * Pi * r^(2)));
```

```
Fe1 := subs(A = 1 , c = 30 , R = 35.67 , t = 0 , Fmag);

F2 := - (0.09461761367 * sqrt(2) * (exp(1)^((71.34 - r) * sqrt
(30) * sqrt(2)) - 1) * exp(1)^(0.01401738156 * sqrt(30) * sqrt(2)
* (35.67 * r - 1)) * sqrt(30))/((exp(1)^(71.31196524 * sqrt(30) *
sqrt(2)) + 1) * r^(2));

### Section 6: figure ###
plot(F2, r = 0.01 .. 1);
```

$$R := 3.567000000 \times 10^{22}$$

$$mp := 1.672621923 \times 10^{-27}$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$r2 := 1.200000000 \times 10^{-15}$$

$$m := 9.109383560 \times 10^{-31}$$

$$c := 299792458$$

$$ec := 1.602176000 \times 10^{-19}$$

$$h := 6.626070150 \times 10^{-34}$$

$$B2 := 6.596177300 \times 10^{12}$$

$$\epsilon 0 := 8.854187813 \times 10^{-12}$$

$$rb := 5.292410932 \times 10^{-11}$$

$$re := 2.820000000 \times 10^{-15}$$

$$rp := 1.200000000 \times 10^{-15}$$

$$r0 := 4.200000000 \times 10^{-15}$$

$$hm := 3.176945500 \times 10^{-27}$$

$$hme := 1.983113649 \times 10^{-8}$$

$$E0 := 5.747437648 \times 10^{-19}$$

$$B1 := 3.910395210$$

$$2.544697800 \times 10^{-19}$$

$$g2 := \frac{2.000000000 \times 10^{-64}}{r^2}$$

$$-\frac{1}{250000000000000000000000000000000000000000000000000000000000000000000000000000 \, r^3}$$

$$-18138.61020$$

$$g1(r) := -\frac{1}{250000000000000000000000000000000000000000000000000000000000000000000000000000000\, r^3}$$

$2.238303432 \times 10^{-29},\ -1.119151716 \times 10^{-29} + 1.938427634 \times 10^{-29}\ \mathrm{I},\ -1.119151716 \times 10^{-29} - 1.938427634 \times 10^{-29}\ \mathrm{I}$

$$r1 := 2.238303432 \times 10^{-29}$$

$2.238303432 \times 10^{-29},\ -1.119151716 \times 10^{-29} + 1.938427634 \times 10^{-29}\ \mathrm{I},\ -1.119151716 \times 10^{-29} - 1.938427634 \times 10^{-29}\ \mathrm{I}$

$$g(7.008690778 \times 10^{-22})$$

$$g1(7.008690778 \times 10^{-22})$$

$$g(2.238303432 \times 10^{-29})$$

$$7.008690778 \times 10^{-22}$$

$$f1 := 4.033609429 \times 10^{-22}$$

$$r1 := \frac{7}{1000000000000000000000}$$

$$\frac{1}{245000000000000000000}$$

$$4.081632653 \times 10^{-22}$$

$$\frac{1}{245000000000000000000}$$

$$omegae := 3.024665977 \times 10^{14}$$

$$omegap := 1.658788001 \times 10^{13}$$

$$0.01990545601$$

$$0.8529558055$$

$$gama := 2.845154315 \times 10^{-9}$$

$$\omega 2 := 4.813905414 \times 10^{13}$$

$$F21 := 15.32298767$$

$$7742.226156$$

$$0.1250696688$$

$$0.1357521327$$

$$Fes := 40.05339802$$

$$FLorentz := 1.937724244 \times 10^{-13}$$

$$Vp := 309286.5961$$

$$499.3757803$$

$$2.303872224 \times 10^{-21}$$

$$1.157407407 \times 10^{-19}$$

$$de := \left(\frac{\partial^2}{\partial t^2} H(r,t)\right)\left(1 + (\mathrm{e})^{-c\,t} A\right) - 2\left(\frac{\partial}{\partial t} H(r,t)\right)(\mathrm{e})^{-c\,t} c\,A + H(r,t)\,(\mathrm{e})^{-c\,t} c^2 A$$
$$- c\left(\frac{\partial^2}{\partial r^2} H(r,t)\right)\left(1 + (\mathrm{e})^{-c\,t} A\right) = 0$$

$$bc := H(2\,R, t) = 0, \mathrm{D}_1(H)\left(\frac{1}{R}, t\right) = R$$

$$H(r,t) = \frac{f_1\left(\frac{t\sqrt{c} + r}{\sqrt{c}}\right) + f_2\left(-\frac{-t\sqrt{c} + r}{\sqrt{c}}\right)}{A\,\mathrm{e}^{-c\,t} + 1}$$

$$Fmag := \left(\sqrt{1 + (\mathrm{e})^{-c\,t} A}\; R\left((\mathrm{e})^{-\frac{2\left(R - \frac{r}{2}\right)\sqrt{c}\sqrt{A}\,(\mathrm{e})^{-\frac{c\,t}{2}}}{\sqrt{1 + (\mathrm{e})^{-c\,t} A}}} - (\mathrm{e})^{\frac{\sqrt{A}\sqrt{c}\,(\mathrm{e})^{-\frac{c\,t}{2}}(2R - r)}{\sqrt{1 + (\mathrm{e})^{-c\,t} A}}}\right)\right) \Bigg/$$
$$\left(4\sqrt{c}\sqrt{A}\left((\mathrm{e})^{-\frac{c\,t\sqrt{1 + (\mathrm{e})^{-c\,t} A}\,R - 4\,(\mathrm{e})^{-\frac{c\,t}{2}}\left(R^2 - \frac{1}{2}\right)\sqrt{A}\sqrt{c}}{2\sqrt{1 + (\mathrm{e})^{-c\,t} A}\,R}} + (\mathrm{e})^{-\frac{c\,t\sqrt{1 + (\mathrm{e})^{-c\,t} A}\,R + 4\,(\mathrm{e})^{-\frac{c\,t}{2}}\left(R^2 - \frac{1}{2}\right)\sqrt{A}\sqrt{c}}{2\sqrt{1 + (\mathrm{e})^{-c\,t} A}\,R}}\right)\pi r^2\right)$$

$$F := \frac{\sqrt{2}\,R\left((\mathrm{e})^{-\left(R - \frac{r}{2}\right)\sqrt{c}\sqrt{2}} - (\mathrm{e})^{\frac{\sqrt{c}\,(2R - r)\sqrt{2}}{2}}\right)}{4\sqrt{c}\left((\mathrm{e})^{\frac{\left(R^2 - \frac{1}{2}\right)\sqrt{c}\sqrt{2}}{R}} + (\mathrm{e})^{-\frac{\left(R^2 - \frac{1}{2}\right)\sqrt{c}\sqrt{2}}{R}}\right)\pi r^2}$$

$$Fmag := -\frac{\sqrt{2}\,R\sinh\left(\left(R - \frac{r}{2}\right)\sqrt{c}\sqrt{2}\right)\operatorname{sech}\left(\sqrt{2}\sqrt{c}\left(R - \frac{1}{2R}\right)\right)}{4\sqrt{c}\,\pi r^2}$$

$$Fe1 :=$$

$$-\frac{0.09461761367\sqrt{2}\sqrt{30}\sinh\left(\left(35.67 - \frac{r}{2}\right)\sqrt{30}\sqrt{2}\right)\operatorname{sech}\left(35.65598262\sqrt{2}\sqrt{30}\right)}{r^2}$$

$$F2 := -\frac{0.09461761367\sqrt{2}\left((e)^{(71.34-r)\sqrt{30}\sqrt{2}}-1\right)(e)^{0.01401738156\sqrt{30}\sqrt{2}(35.67r-1)}\sqrt{30}}{\left((e)^{71.31196524\sqrt{2}\sqrt{30}}+1\right)r^2}$$

